

Interpretable Machine Learning for ICU Mortality Prediction

Deep learning has emerged as one of the most powerful machine learning techniques of the last decade. Neural networks with many layers can now learn complex patterns directly from raw data such as images, audio and text. In the medical domain, deep learning models have achieved expert-level performance on tasks like classifying skin conditions, detecting diabetic retinopathy in eye scans, and predicting patient deterioration from electronic health records.

Despite this progress, the adoption of deep learning in clinical practice remains limited. A key obstacle is the lack of interpretability: clinicians hesitate to trust predictions from models that cannot explain their reasoning. Consequently, research on explainable artificial intelligence for healthcare has grown rapidly in recent years, producing techniques such as attention maps, saliency methods and concept activation vectors.

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In this work we propose an interpretable pipeline for hospital mortality prediction. Our approach combines a gradient boosted tree ensemble with post-hoc explanation via SHAP values. We evaluate the pipeline on the MIMIC-IV intensive care database, which contains de-identified records of more than fifty thousand ICU admissions. Our model achieves an area under the ROC curve of 0.85, exceeding previously reported baselines while remaining fully explainable at the patient level.

Our results suggest that interpretable machine learning can match the accuracy of black-box deep networks on tabular clinical data. We argue that future work should focus on human-in-the-loop evaluation, where explanations are assessed not only by fidelity metrics but by whether they help clinicians make better and faster decisions at the bedside.